THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** handpicking	Used when pieces are large.
2	**(c)** condensation	Cooling vapour.
3	**(c)** distilled water	Only one substance.
4	**(c)** winnowing	Wind blows away light husk.
5	**(a)** filtration	Sand is insoluble.
6	**(a)** air	Gases mixed.
7	**(c)** threshing	Threshing frees grains.
8	**(a)** evaporation	Salt is dissolved.
9	**(b)** wheat and husk	Different weights.
10	**(a)** sieving	Different particle sizes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two or more substances mixed together (1); Salt and sand (1).	1+1
12	By evaporation of water (1); Salt crystals remain (1).	1+1
13	Sieving flour (1); Straining tea (1).	1+1
14	To remove harmful or unwanted parts (1); To get useful substances (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Allow to stand so sand settles (1); Pour off water (decantation) (1); Filter the rest (1).	1+1+1
16	Boil or cool milk (1); Cream forms a layer (1); Remove it or churn (1).	1+1+1
17	Higher temperature dissolves more (1); Dissolves faster (1); Cooling may cause crystals to appear (1).	1+1+1
	OR: Add water to dissolve salt (1); Filter to remove sand (1); Evaporate filtrate to get salt (1).	1+1+1
18	Move a magnet over the mixture (1); Nails stick (1); Rice remains (1).	1+1+1
	OR: Particles are tiny (1); Too many particles (1); Dissolving is better (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Filter the mixture (1); Sand remains as residue (1); Filtrate has salt (1); Evaporate the filtrate (1); Salt crystals remain (1).	1+1+1+1+1
	OR: Heat sea water (1); Water evaporates (1); Condense vapour on a cool surface (1); Collect drops (1); Salt remains (1).	1+1+1+1+1
20	Keep muddy water still (1); Mud settles: sedimentation (1); Pour off clear water: decantation (1); Pass through filter paper (1); Boil for safe drinking (1).	1+1+1+1+1
	OR: Sieving (1); Wheat flour (1); Sand and gravel (1); Tea leaves (1); Hole size decides (1).	1+1+1+1+1

Mistakes to avoid:

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** magnet	Iron is magnetic.
2	**(a)** wheat flour and bran	Particle size differs.
3	**(a)** sedimentation	Sediment collects at the bottom.
4	**(d)** churning	Spinning separates cream.
5	**(b)** threshing	Beating stalks.
6	**(c)** decantation	Gentle pouring off.
7	**(b)** strainer	Strainer = filter.
8	**(b)** sedimentation, decantation, filtration	Standard purification.
9	**(a)** filtration	The residue stays on the paper.
10	**(d)** sedimentation	Settles under gravity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Separating husk from grains using wind (1); Lighter chaff blows away (1).	1+1
12	Spinning milk to separate cream (1); Cream floats up (1).	1+1
13	Settling of heavy particles (1); Pouring off the clear liquid (1).	1+1
14	Using a magnet (1); Iron sticks to it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sedimentation (1); Decantation (1); Filtration and boiling (1).	1+1+1
16	Handpicking (1); Sieving (1); Winnowing and magnet (1).	1+1+1
17	Sea water in shallow fields (1); Sun evaporates water (1); Salt left behind (1).	1+1+1
	OR: Substances have different properties (1); Size, weight or solubility differ (1); Method matches the property (1).	1+1+1
18	Sedimentation (1); Decantation (1); Filtration and evaporation (1).	1+1+1
	OR: Boil tea leaves (1); Pour through a strainer (1); Leaves are the residue (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Handpicking: stones from rice (1); Winnowing: husk from wheat (1); Sieving: flour (1); Threshing: grain from stalks (1); Choice by size, weight (1).	1+1+1+1+1
	OR: Allow it to stand undisturbed (1); Oil is lighter and floats as a separate layer (1); Pour or spoon off the oil layer carefully (1); This is decantation (1); Water is left behind (1).	1+1+1+1+1
20	Evaporation: liquid to vapour (1); Salt left (1); Condensation: vapour to liquid (1); Pure water obtained by distillation (1); Both use heat or cooling (1).	1+1+1+1+1
	OR: Solid-solid: sieving (1); Solid-liquid: filtration (1); Liquid-liquid: decantation (1); Dissolved solid: evaporation (1); Based on properties (1).	1+1+1+1+1

Mistakes to avoid:

- Choosing filtration for salt in water; dissolved salt passes through the filter, so **evaporation** is needed.
- Mixing **sedimentation** (settling) with **decantation** (pouring off the clear liquid).
- Mixing **threshing** (beating stalks to free grain) with winnowing (blowing away husk).
- Calling the liquid that passes through filter paper the residue; it is the **filtrate**.
- Using a magnet to separate sand or salt; only magnetic substances like iron are attracted.
- Believing that filtered or settled water is always safe to drink.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Methods of Separation in Everyday Life** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** saturation	Maximum limit reached.
2	**(a)** residue	It stays behind.
3	**(b)** solute	Salt in salt water.
4	**(d)** settles at the bottom	No more dissolves.
5	**(a)** filtrate	It is collected below.
6	**(c)** solvent	Water is the common solvent.
7	**(b)** more solute	Higher temperature increases solubility.
8	**(c)** evaporation	Water evaporates leaving salt.
9	**(d)** mixture	Mixed in any proportion.
10	**(a)** water changes to vapour and salt remains	Salt does not evaporate.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Beating harvested stalks to remove grains (1); Used by farmers (1).	1+1
12	Separating insoluble solids from liquids (1); Water purification, tea strainer (1).	1+1
13	Solution that cannot dissolve more solute (1); At that temperature (1).	1+1
14	Solid left on filter paper (1); Remains behind (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wind blows away light husk (1); Heavy grains fall down (1); Gets clean grain (1).	1+1+1
16	Holes let liquid pass (1); Leaves remain (1); Acts as a filter (1).	1+1+1
17	Flour (1); Sand at construction (1); Tea leaves (1).	1+1+1
	OR: Add salt in water (1); Stir and add till no more dissolves (1); Extra salt settles (1).	1+1+1
18	Salt is dissolved (1); It passes through filter paper (1); Evaporation is needed (1).	1+1+1
	OR: Add water (1); Sugar dissolves (1); Filter and evaporate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solution: uniform mixture (1); Solute: dissolved substance (1); Solvent: liquid in which it dissolves (1); Saturated: no more can dissolve (1); Example: salt in water (1).	1+1+1+1+1
	OR: Sediment filter removes dirt (1); Carbon filter removes smell (1); Membrane or UV for germs (1); Safe water (1); Steps of separation (1).	1+1+1+1+1
20	Works for iron (1); Mix of iron and sand (1); Pass magnet (1); Iron clings (1); Sand remains (1).	1+1+1+1+1
	OR: Handpick stones (1); Wash rice (1); Decant dirty water (1); Repeat (1); Clean rice remains (1).	1+1+1+1+1

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